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import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Step 1: Upload the input image
uploaded = files.upload()

# Get the uploaded file name
image_path = next(iter(uploaded))

# Step 2: Read the image
image = cv2.imread(image_path)

# Check if image was loaded
if image is None:
    print("Error: Image could not be loaded.")
else:

    # Step 3: High-Boost Filtering
    def high_boost_filter(image, boost_factor):
        kernel_size = 3

        # Average filter kernel
        kernel = np.ones(
            (kernel_size, kernel_size),
            dtype=np.float32
        ) / (kernel_size ** 2)

        # Blur the image
        blur_image = cv2.filter2D(image, -1, kernel)

        # Convert to float to avoid overflow
        image_float = image.astype(np.float32)
        blur_float = blur_image.astype(np.float32)

        # High-boost mask
```

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sharpened = image_float + boost_factor * (image_float - blur_float)

# Keep pixel values between 0 and 255
sharpened = np.clip(sharpened, 0, 255)

return sharpened.astype(np.uint8)

# Step 4: Apply high-boost filter
sharpened_image = high_boost_filter(image, 1.5)

# Step 5: Save the output image
output_path = "sharpened_image.jpg"
cv2.imwrite(output_path, sharpened_image)

# Step 6: Display original and sharpened images
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(sharpened_image, cv2.COLOR_BGR2RGB))
plt.title("High-Boost Filtered Image")
plt.axis("off")

plt.show()

# Step 7: Download the output
files.download(output_path)
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Choose Files cvvv5.png

cvvv5.png(image/png) - 290257 bytes, last modified: 7/22/2026 - 100% done
Saving cvvv5.png to cvvv5.png

Original Image



High-Boost Filtered Image



